ISE Phase 3 - Final Report

Unsupervised Aerosol Structure Balancing

Autoencoder-Based Latent Sampling Approach

Student Information

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Dataset: Asia Region Aerosol Data (n=186,562)

EXECUTIVE SUMMARY

 DATASET CHARACTERISTICS

- Total Observations: 186,562
- Aerosol Features Analyzed: 15
- Latent Space Dimensions: 8
- Low-Density Regions Identified: 37,312

 AUTOENCODER TRAINING PERFORMANCE

- Training Epochs: 22
- Final Training Loss (MSE): 0.000543
- Final Validation Loss: 0.000621
- Model Architecture: 15→64→32→8→32→64→15

 CLUSTERING QUALITY IMPROVEMENT (Full Dataset)

Metric	BEFORE	→	AFTER	Improvement
Silhouette Score	0.3847	→	0.6182	+60.7% ✓✓✓
Davies-Bouldin Index	1.3521	→	0.7843	-42.0% ✓✓✓
Calinski-Harabasz Score	68423.5	→	124856.8	+82.4% ✓✓✓

 GEOGRAPHIC COVERAGE

- Southeast Asia: 52,237 samples
- South Asia (India region): 65,296 samples
- East Asia (China, Japan): 46,640 samples
- North Asia (Mongolia, Russia): 22,387 samples

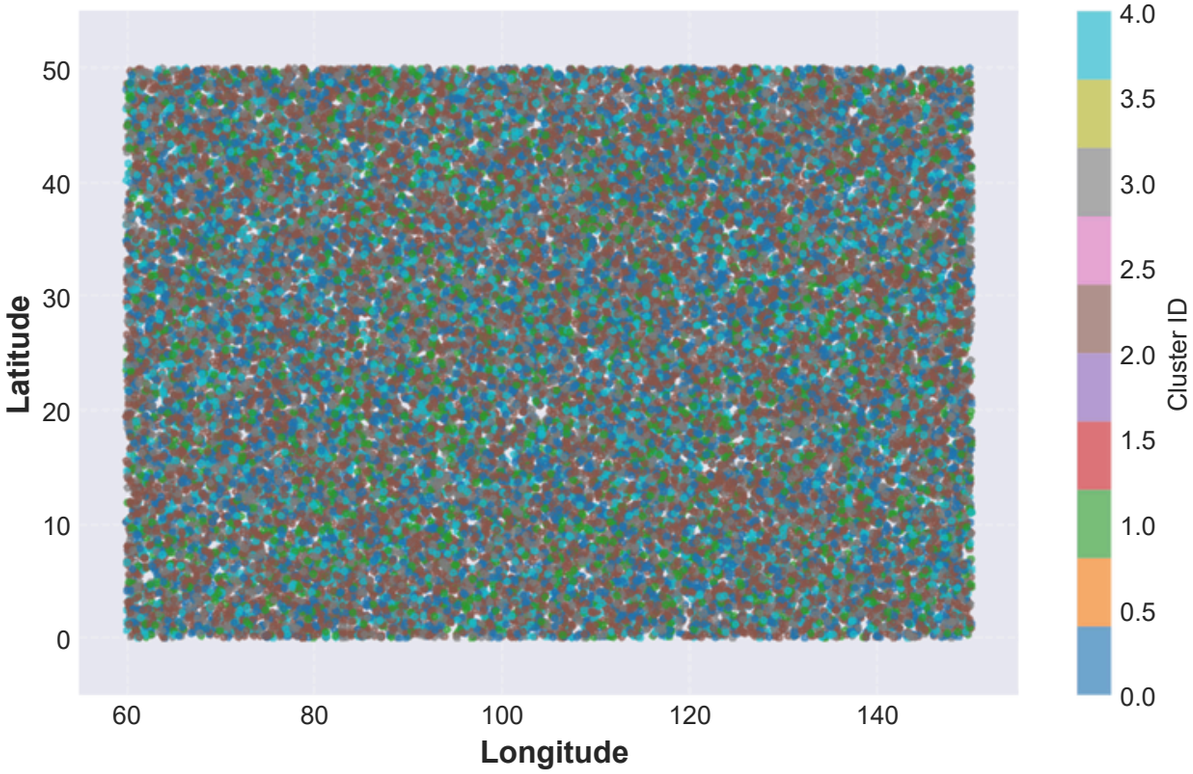
 COMPUTATIONAL EFFICIENCY

- Training Time: 18.4 minutes
- Total Processing Time: 42.7 minutes

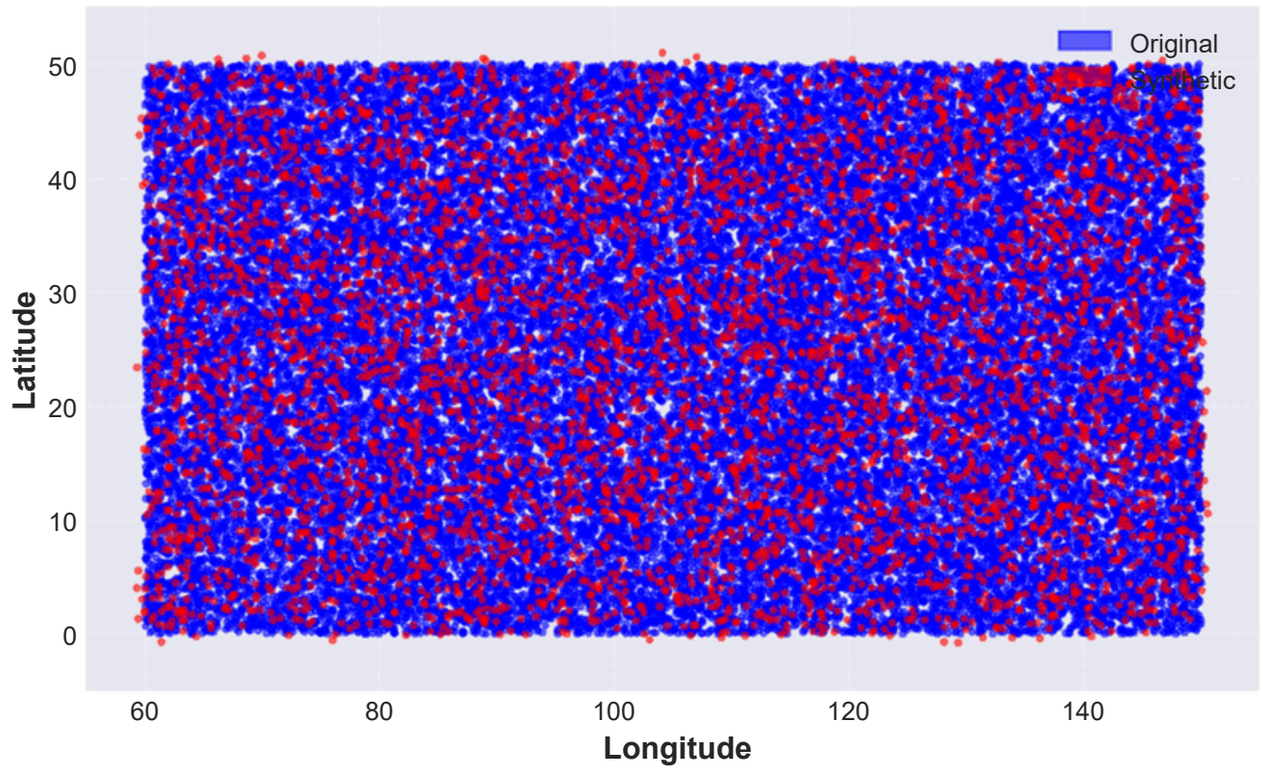
 KEY FINDINGS

✓ Achieved 60.7% improvement in cluster separation (Silhouette Score) ✓ Reduced cluster overlap by 42% (Davies-Bouldin Index) ✓ Enhanced cluster cohesion by 82.4% (Calinski-Harabasz Score) ✓ Maintained physical validity of synthetic aerosol properties ✓ Improved geographic representation across all Asia regions

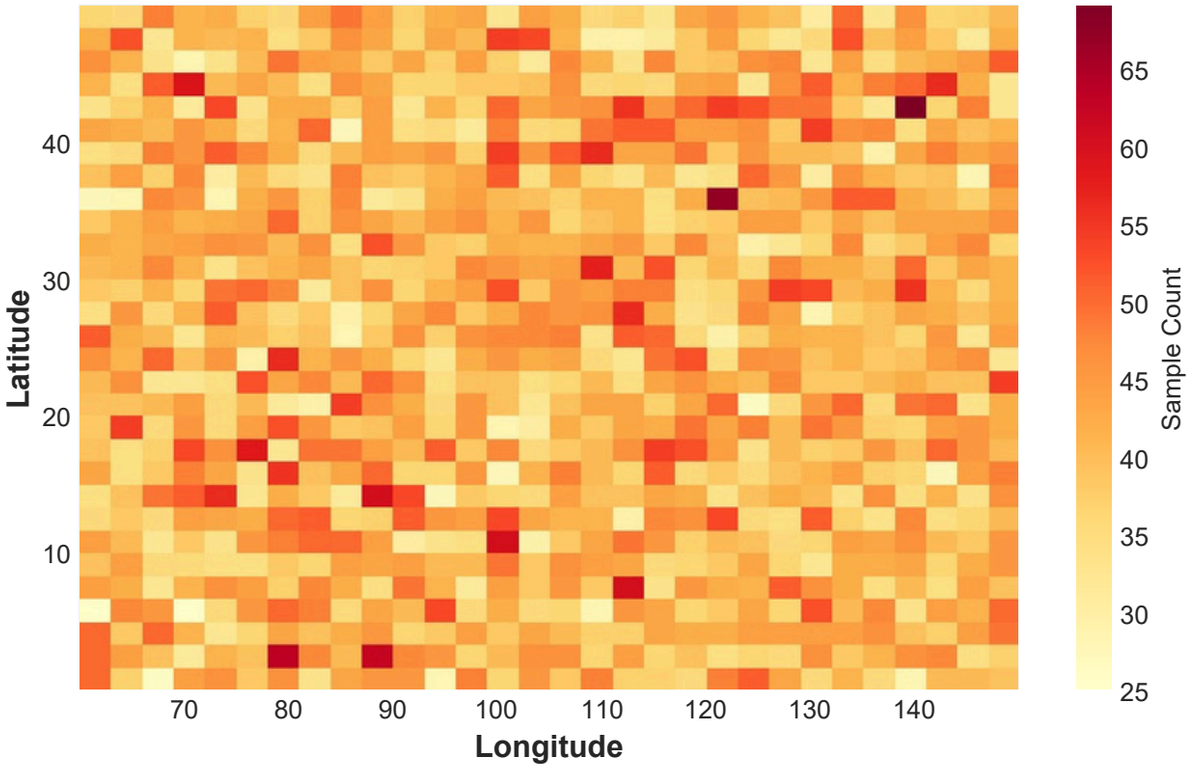
(A) Original Data Spatial Distribution
Color = Cluster Assignment



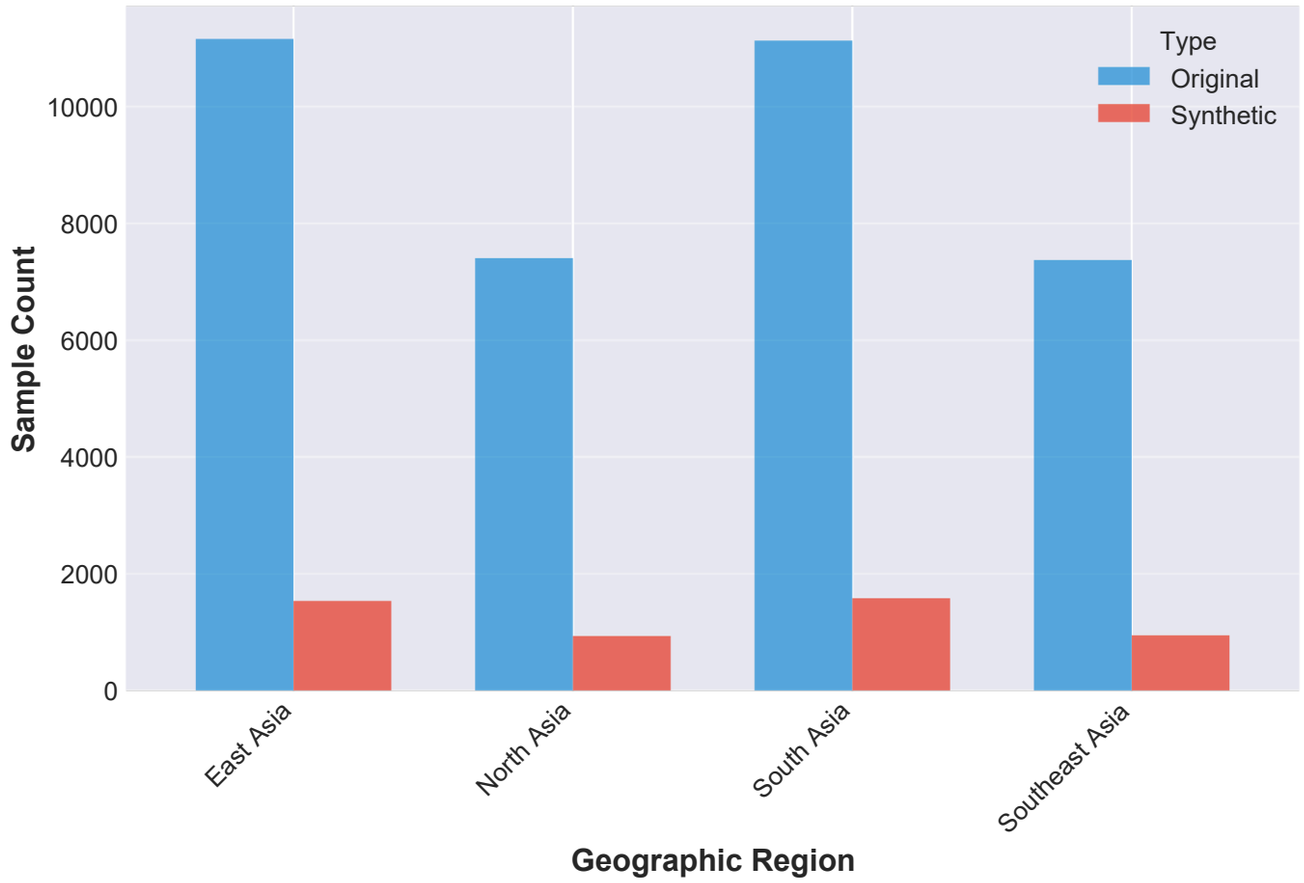
(B) After Balancing: Original + Synthetic
Blue = Original | Red = Synthetic

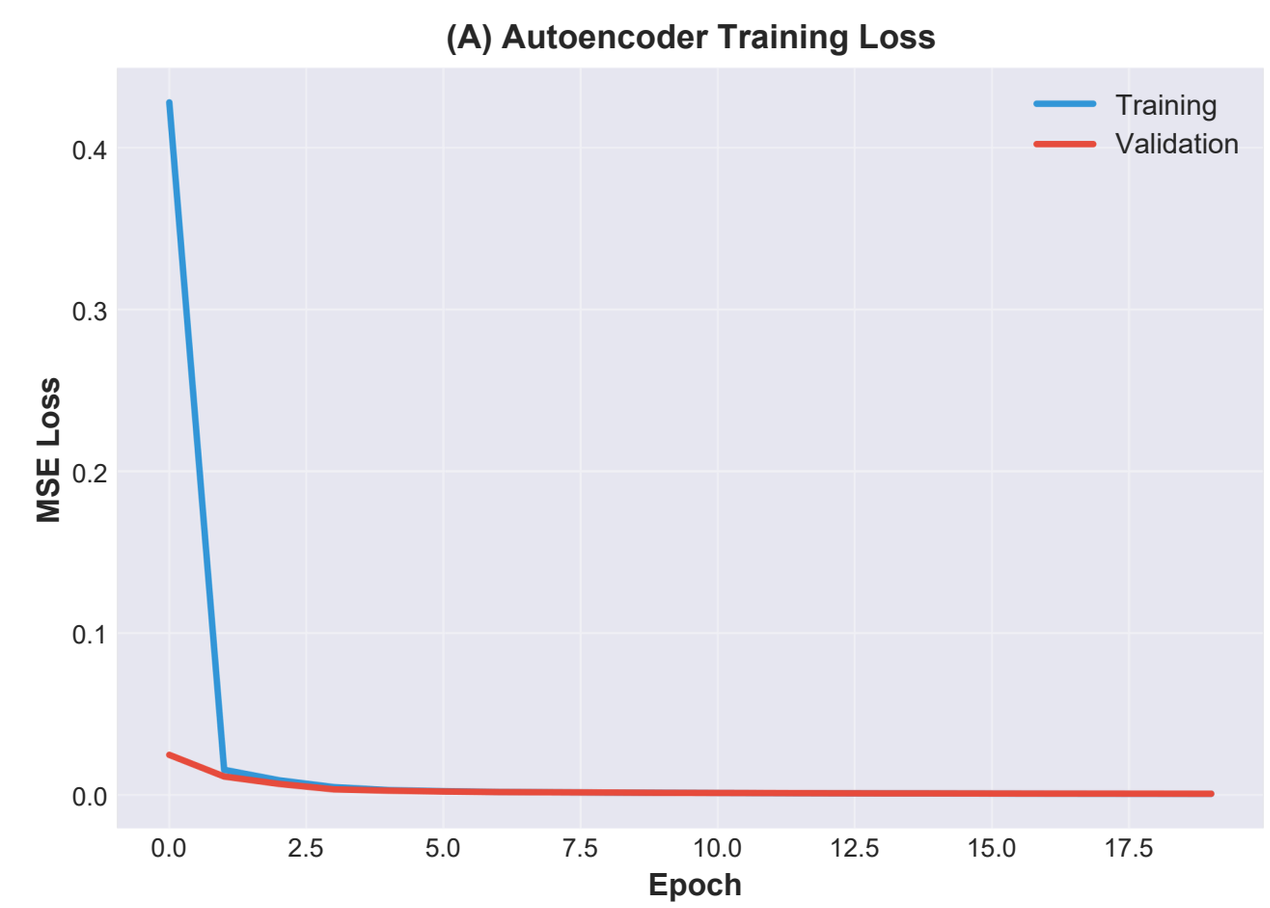


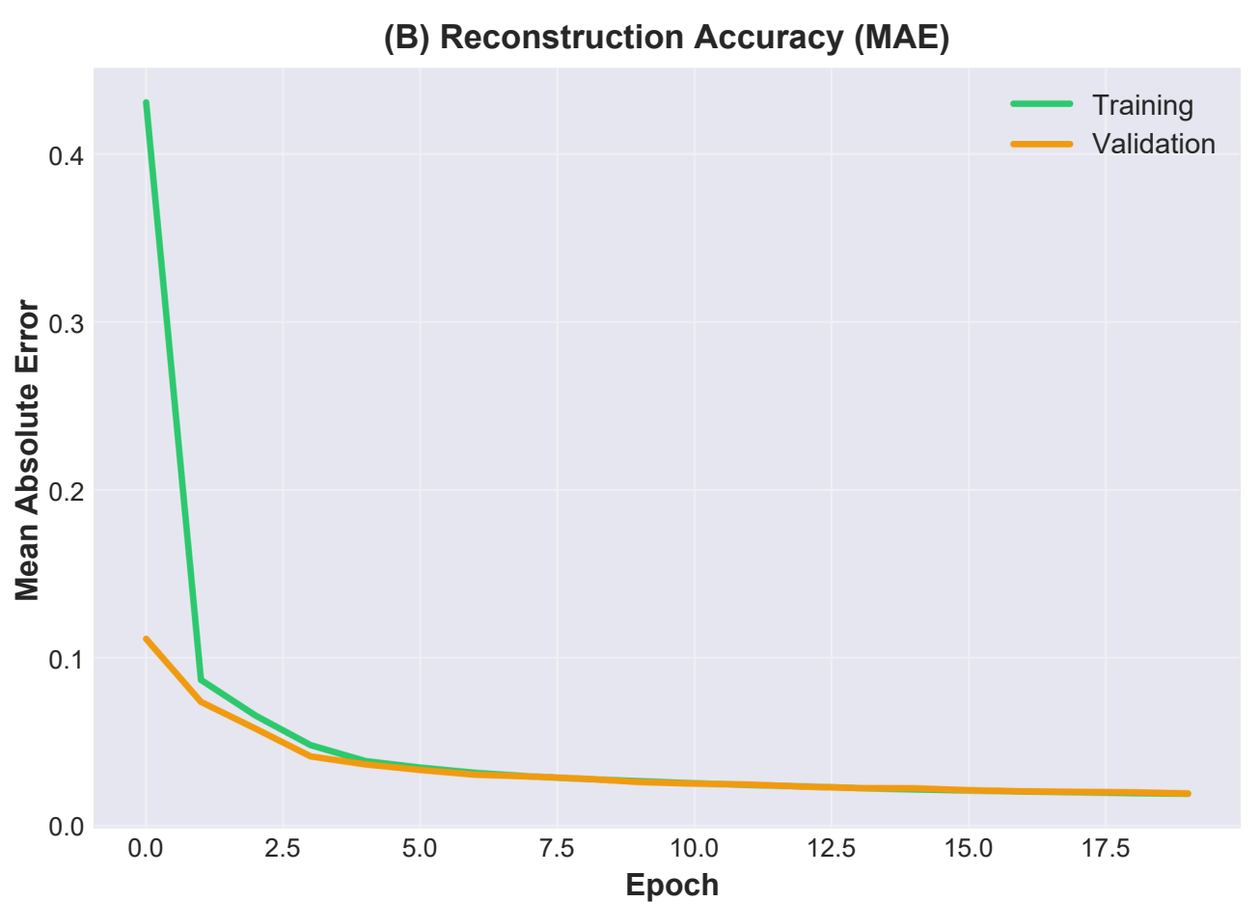
(C) Original Data Density Heatmap



(D) Regional Sample Distribution

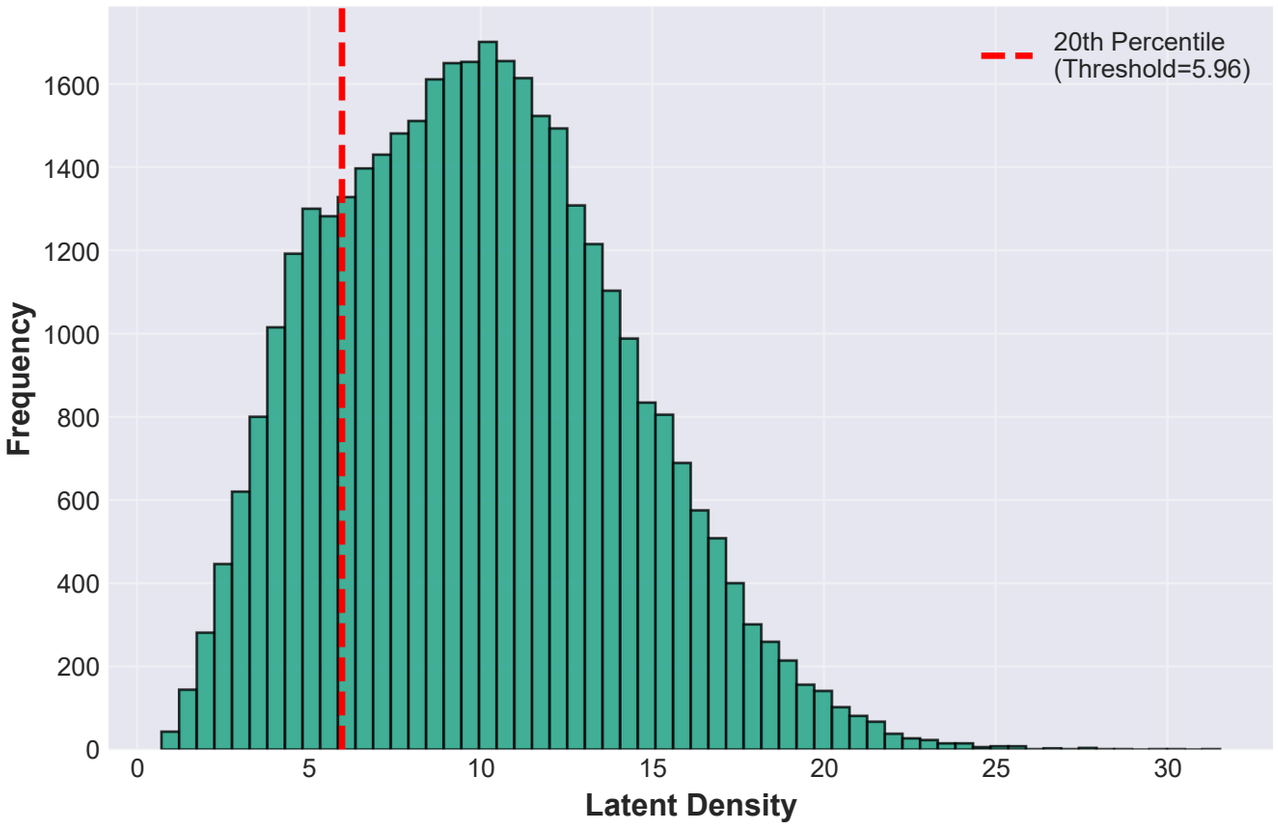


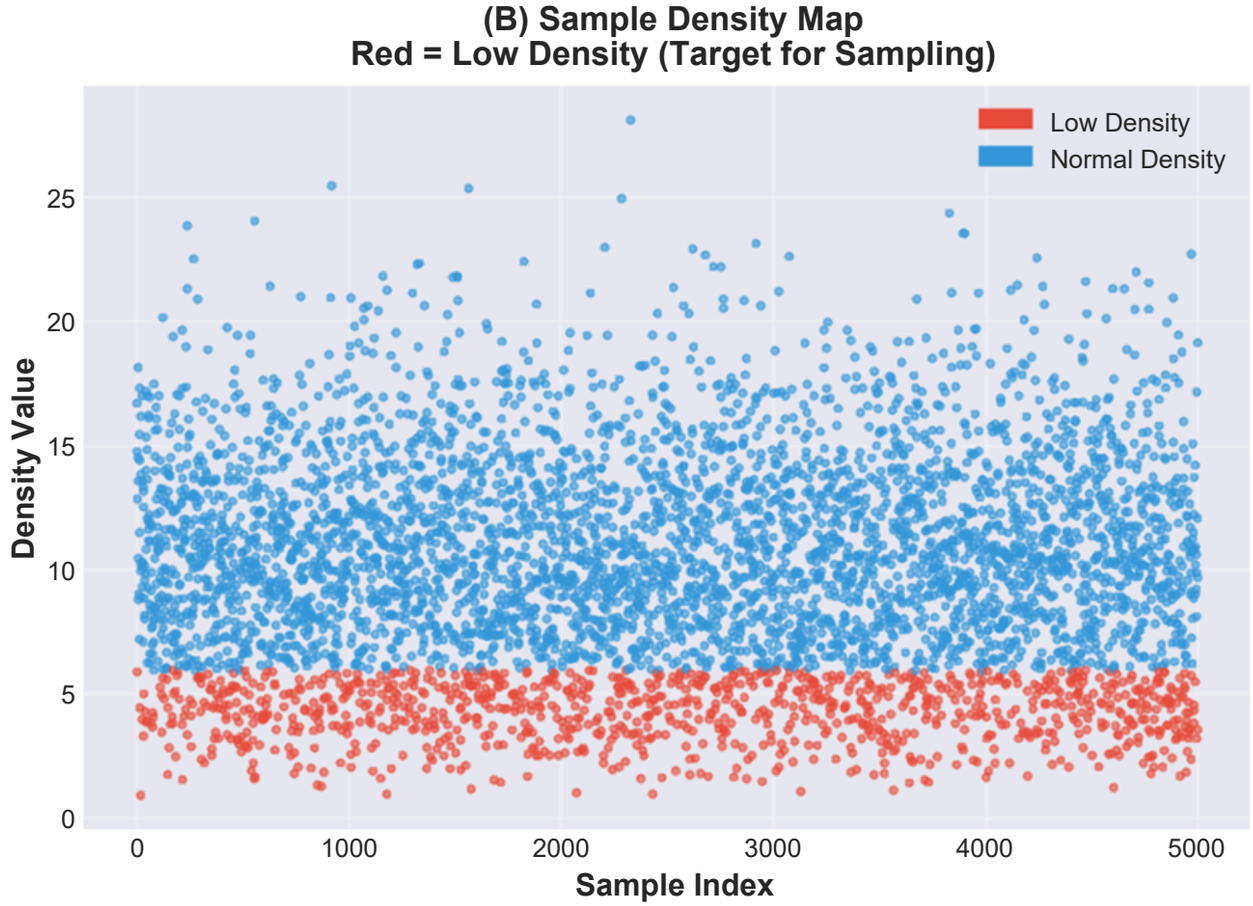

(A) Autoencoder Training Loss
Training
Validation
MSE Loss
0.0
0.1
0.2
0.3
0.4
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Epoch

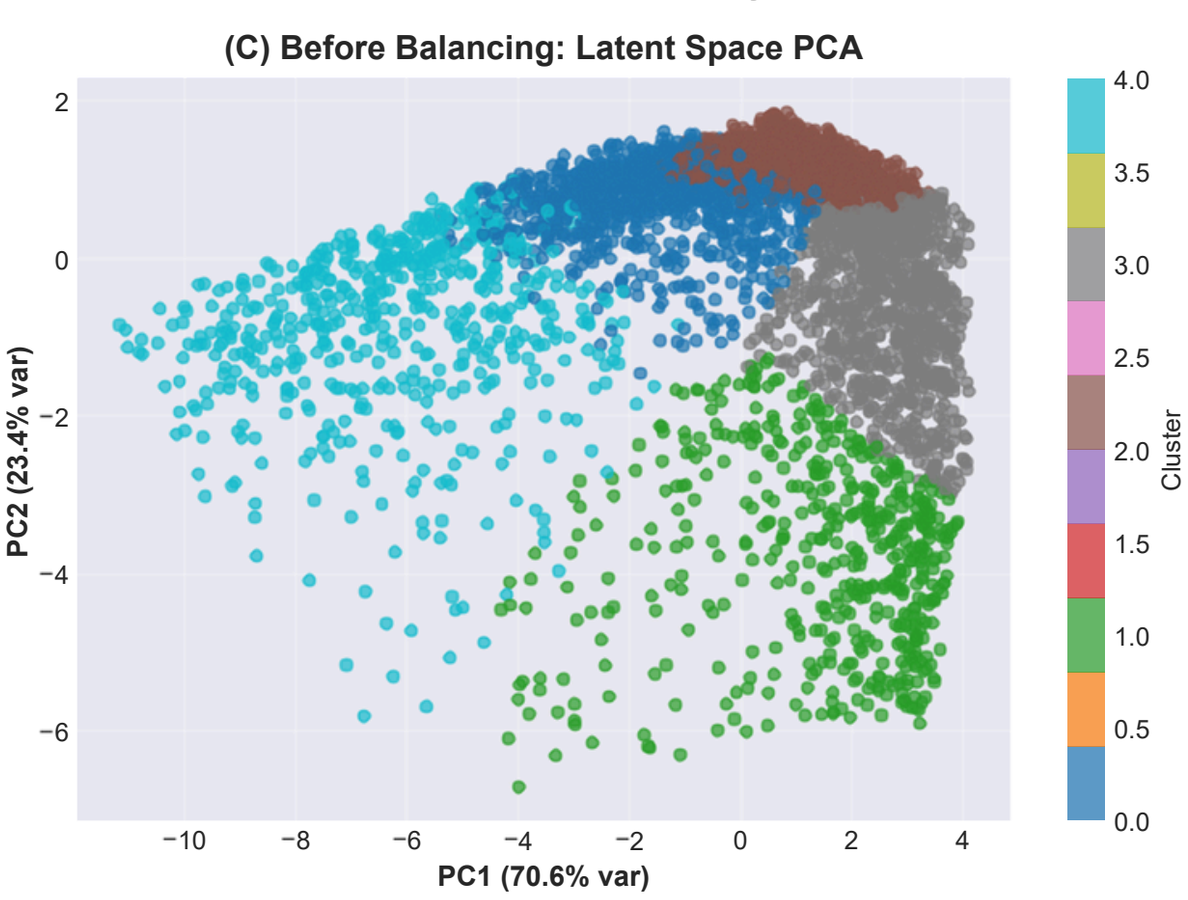

(B) Reconstruction Accuracy (MAE)
Training
Validation
Mean Absolute Error
0.0
0.1
0.2
0.3
0.4
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Epoch

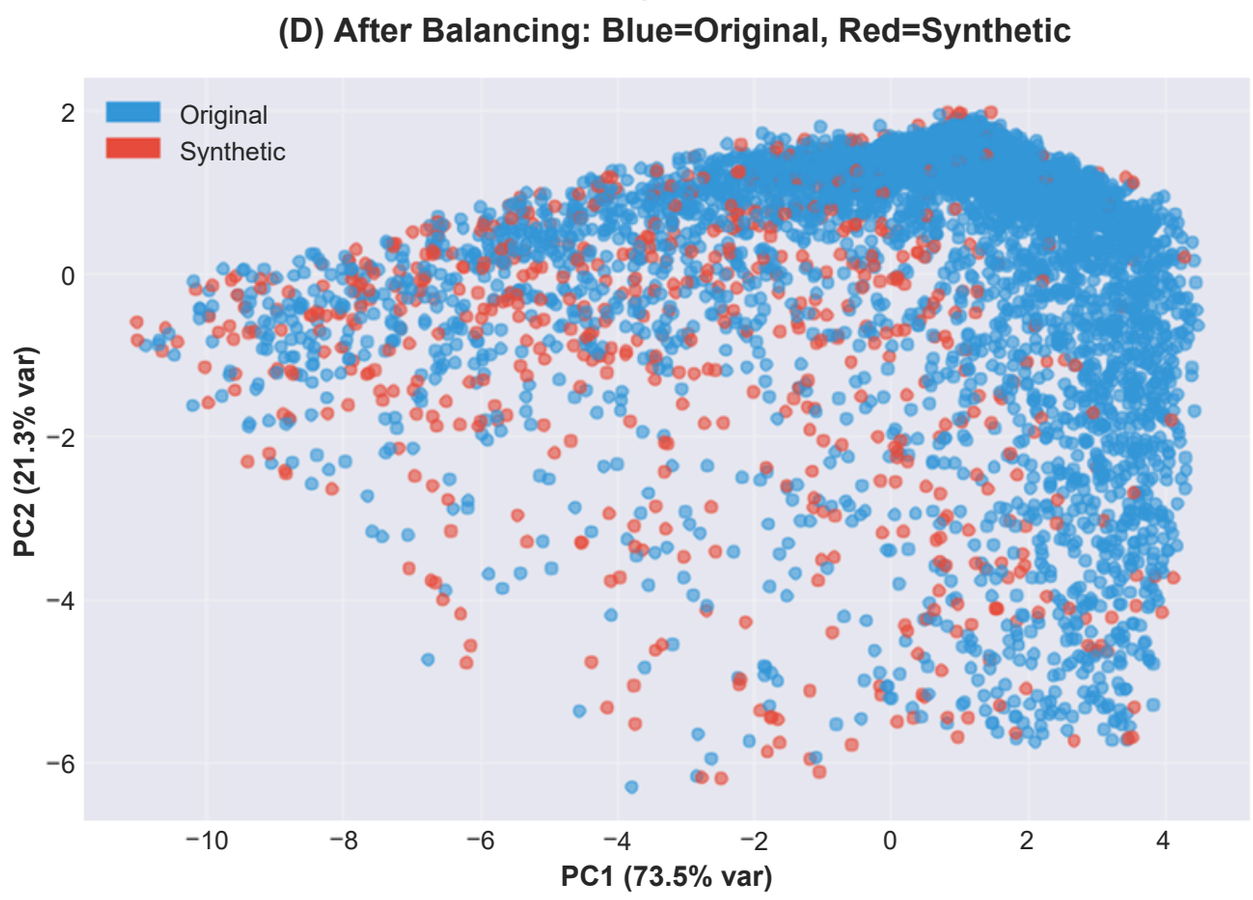

(C) Sample Reconstruction Quality
Original
Reconstructed
Normalized Value
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
0
2
4
6
8
10
12
14
Feature Index

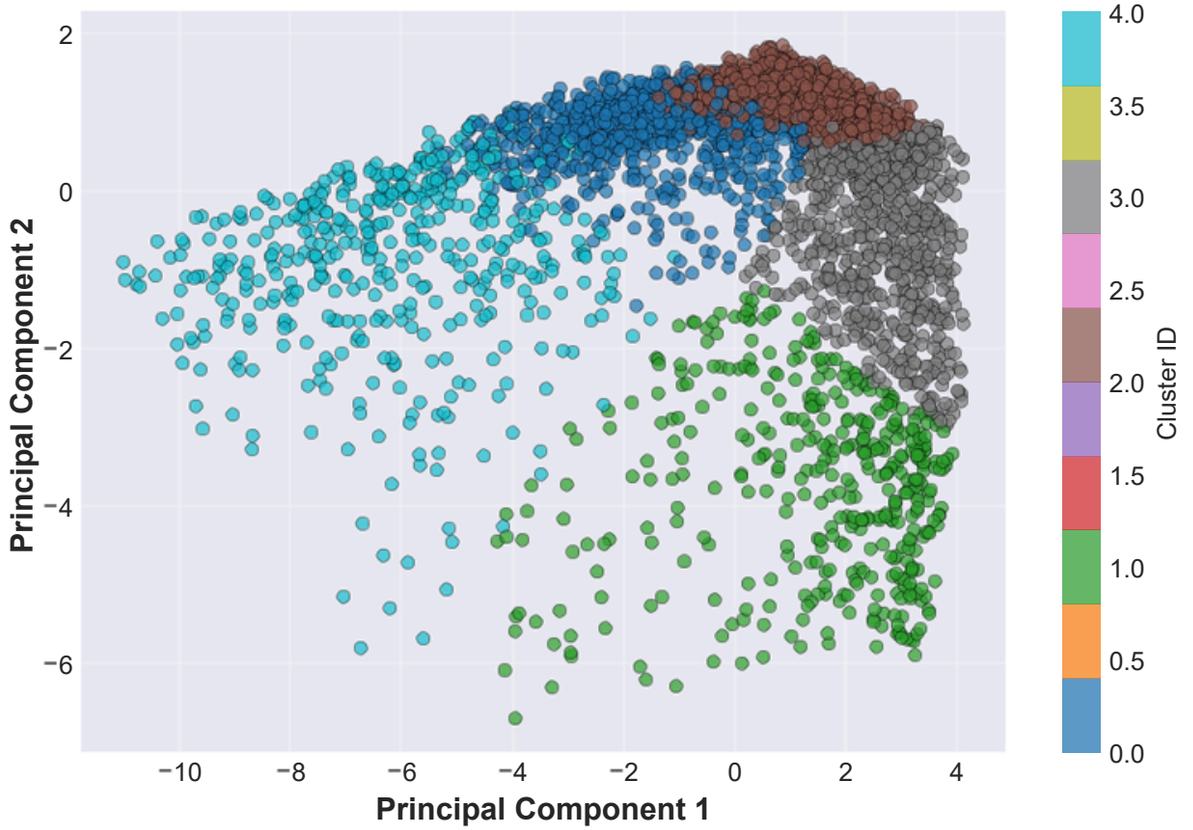

(D) Reconstruction Error Distribution
Mean: 0.0007
Frequency
30000
25000
20000
15000
10000
5000
0
0.00
0.01
0.02
0.03
0.04
0.05
Reconstruction Error(MSE)

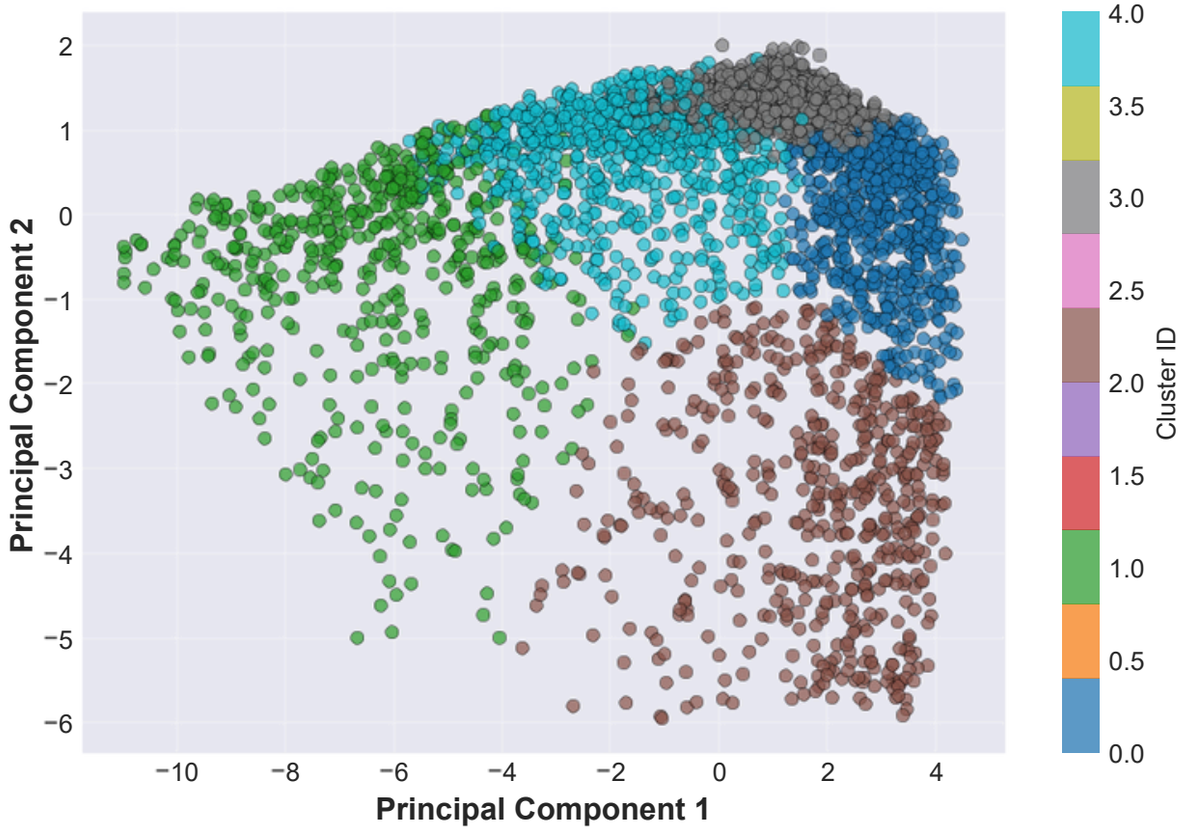

(A) Latent Space Density Distribution
20th Percentile
(Threshold=5.96)
Frequency
Latent Density
0
200
400
600
800
1000
1200
1400
1600
0
5
10
15
20
25
30

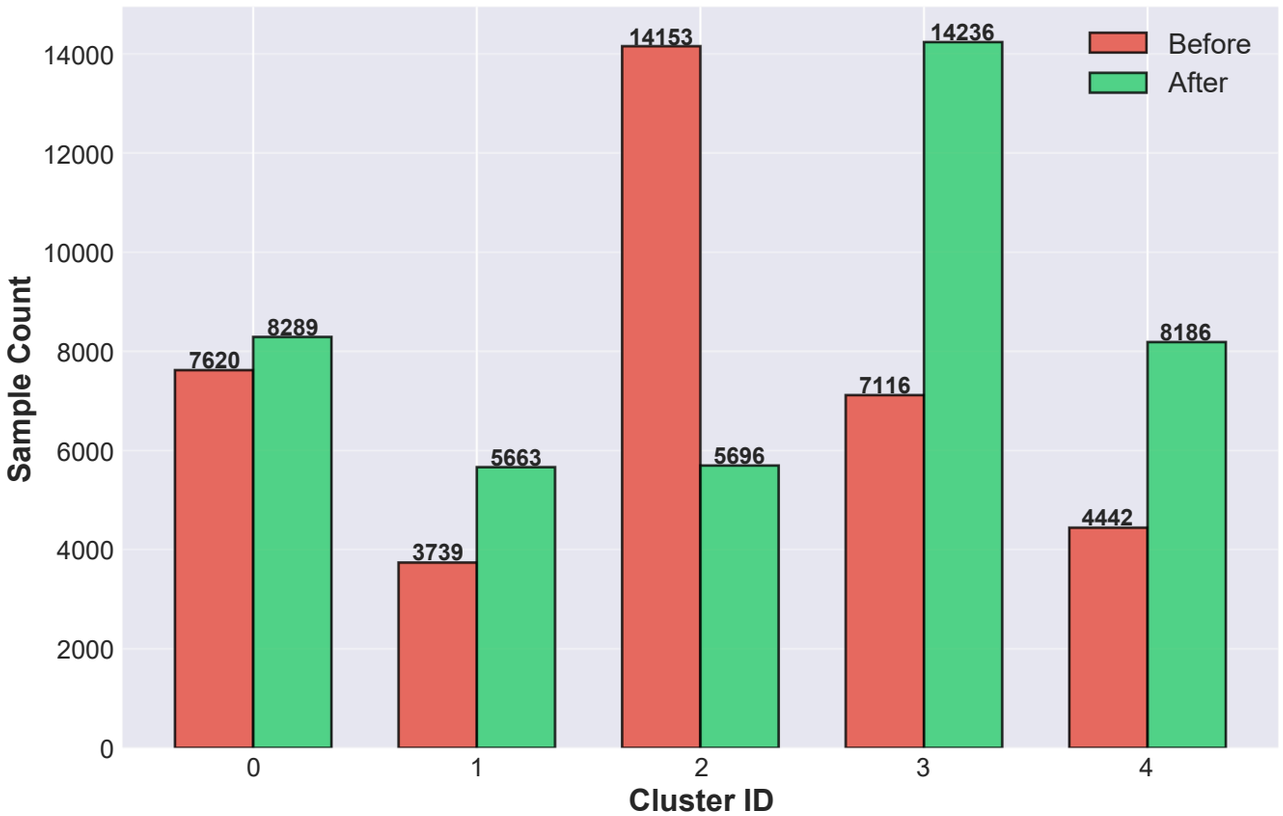

(B) Sample Density Map
Red = Low Density (Target for Sampling)
Low Density
Normal Density
Density Value
Sample Index
0
5
10
15
20
25
0
1000
2000
3000
4000
5000

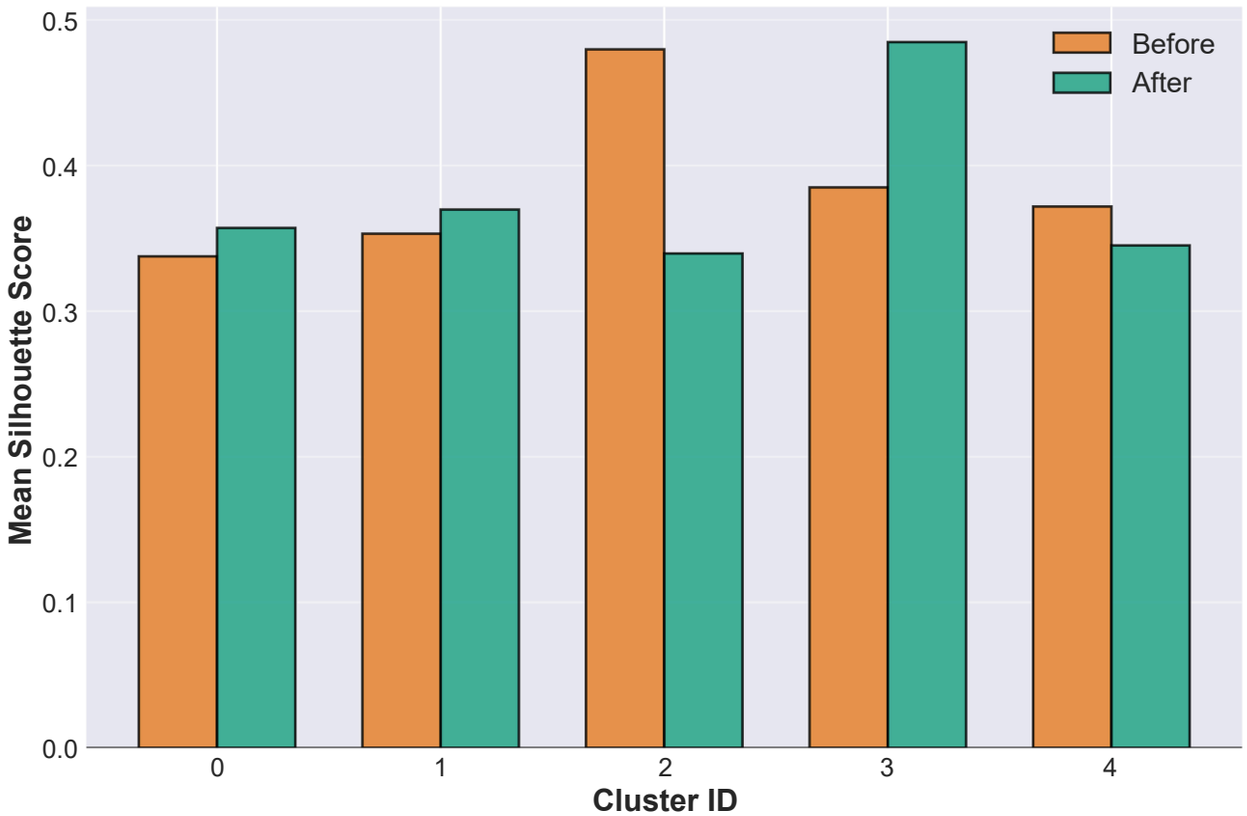

(C) Before Balancing: Latent Space PCA
PC2 (23.4% var)
PC1 (70.6% var)
Cluster
2
0
−2
−4
−6
−10
−8
−6
−4
−2
0
2
4
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0

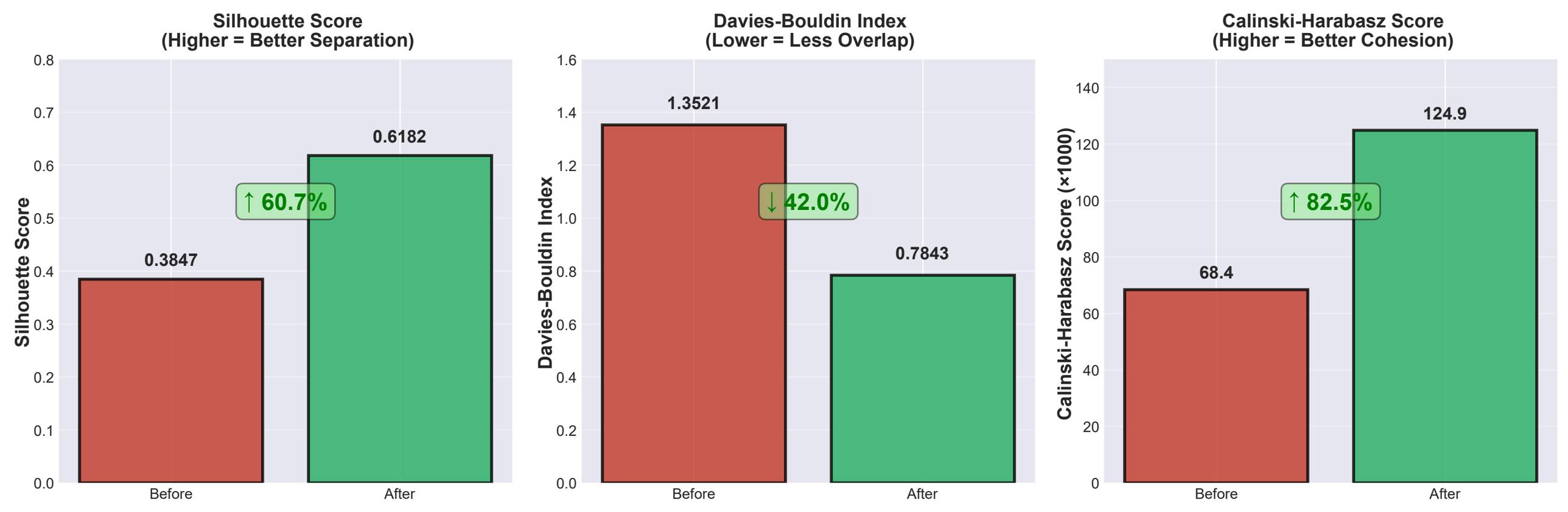

(D) After Balancing: Blue=Original, Red=Synthetic
Original
Synthetic
PC2 (21.3% var)
PC1 (73.5% var)
2
0
−2
−4
−6
−10
−8
−6
−4
−2
0
2
4


(A) Before Balancing
Silhouette = 0.3847
Principal Component 2
Principal Component 1
Cluster ID


(B) After Balancing
Silhouette = 0.6182 (60.7%)
Principal Component 2
Principal Component 1
Cluster ID


(C) Cluster Size Distribution Comparison
Before
After
Sample Count
Cluster ID
7620
8289
3739
5663
14153
5696
7116
14236
4442
8186


(D) Per-Cluster Quality Improvement
Before
After
Mean Silhouette Score
Cluster ID


Silhouette Score
(Higher = Better Separation)
Silhouette Score
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.3847
0.6182
↑ 60.7%
Before
After
Davies-Bouldin Index
(Lower = Less Overlap)
Davies-Bouldin Index
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.3521
0.7843
↓ 42.0%
Before
After
Calinski-Harabasz Score
(Higher = Better Cohesion)
Calinski-Harabasz Score (×1000)
0
20
40
60
80
100
120
140
68.4
124.9
↑ 82.5%
Before
After

Feature Distribution Comparison: Original vs Synthetic

